VIX Tupiniquim I: Construction of a Synthetic Financial Stress Indicator via PCA and Linear Regularization

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Abstract

This exercise proposes the construction and validation of the "VIX Tupiniquim", a synthetic indicator aimed at capturing systemic risk and Brazilian domestic financial stress. To deal with the challenges inherent to the high dimensionality of real macroeconomic and financial data, a hybrid Machine Learning econometric strategy is employed. Initially, Principal Component Analysis (PCA) is applied for the extraction of informational latent factors. Next, linear regularization techniques (LASSO, Ridge, and Elastic Net) are used for the parsimonious selection of variables and robust estimation. The results demonstrate the effectiveness of the approach in the robust measurement of market "fear", offering an alternative adapted to the idiosyncrasies of the Brazilian economy.

Keywords: Systemic Risk, High Dimensionality, Linear Regularization, PCA, Brazilian "Fear" Index.

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1 Introduction

The timely measurement of systemic risk and volatility in domestic financial markets represents a recurring challenge for analysts and monetary authorities for decision-making. Although the VIX index (calculated by CBOE) serves as a global thermometer of "fear", its direct application in emerging economies, such as Brazil, often tends to capture only partially the peculiarities of local shocks, sovereign risk premiums, and exchange rate frictions.

This exercise proposes the construction of the *VIX Tupiniquim*, a synthetic indicator of financial stress structured from a wide range of national macroeconomic and financial variables. To deal with the high dimensionality and noise inherent to this data panel, a hybrid quantitative approach is employed, combining dimensionality reduction by Principal Component Analysis (PCA) and linear regularization techniques from Machine Learning (LASSO, Ridge, and Elastic Net). The objective is to identify the most informative components and parsimoniously select the most informative variables for diagnosing domestic systemic stress.

2 Methodologies

To construct the VIX Tupiniquim and deal with a large amount of variables without falling into the trap of overfitting, two complementary approaches were used: one to condense information and another to select the truly important factors.

2.1 Principal Component Analysis (PCA)

When there are dozens of indicators pointing in similar directions (exchange rate, country risk, interest rates, volatility), PCA acts as a "statistical synthesis". Instead of looking at each series individually and generating noise, the technique extracts a central axis — a synthesis factor — that carries most of the information and energy from the entire original dataset.

Mathematically, given a set of standardized variables $X_1, X_2, \dots, X_p$, the first principal component (PC_1) is constructed as a linear combination that maximizes the explained variance:

$$PC_1 = w_1 X_1 + w_2 X_2 + \dots + w_p X_p \tag{1}$$

where the weights w_j (eigenvectors) determine the degree of contribution of each original variable to the formation of the stress thermometer.

2.2 Linear Regularization (LASSO, Ridge, and Elastic Net)

Since not all variables in the panel have the same weight in explaining market stress, linear regularization methods were used to "clean" the model. The central idea is to penalize the size of the coefficients to prevent irrelevant variables from causing predictive noise.

1. **LASSO (*Least Absolute Shrinkage and Selection Operator*):** Adds a penalty based on the sum of the absolute values of the coefficients (L_1 norm). Its great advantage is that it can completely zero out the coefficient of irrelevant variables, performing an automatic feature selection:

$$\min_{\beta} \left(\frac{1}{2T} ||y - X\beta||_2^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right) \tag{2}$$

2. **Ridge (Ridge Regression):** The term "ridge" refers to the original algebraic formulation that stabilized the data matrix. The method adds a penalty based on the square of the coefficients (L_2 norm). It reduces the impact of collinear variables by shrinking their coefficients toward zero, but without eliminating them completely:

$$\min_{\beta} \left(\frac{1}{2T} ||y - X\beta||_2^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right) \tag{3}$$

3. **Elastic Net:** Combines simultaneously the penalties of LASSO and Ridge (mixing the L_1 and L_2 norms). It is ideal for scenarios with many correlated variables, as it inherits the selection capability of LASSO with the grouping stability of Ridge:

$$\min_{\beta} \left(\frac{1}{2T} ||y - X\beta||_2^2 + \lambda_1 \sum_{j=1}^{p} |\beta_j| + \lambda_2 \sum_{j=1}^{p} \beta_j^2 \right) \tag{4}$$

Where λ (or λ_1, λ_2) represents the regularization parameter that controls the rigor of the penalty applied by the model. In the following section, we will test the practical performance of these approaches.

3 Data Used

For the construction of the *VIX Tupiniquim*, variables of high economic relevance from the Brazilian scenario were collected. These variables have monthly and daily periodicities, with the latter duly converted to a monthly basis.

All series were collected in their original format, preserving native adjustment methods. For the validation of the exercise, the structural stochastic filter *Seasonal-Trend decomposition using Loess* (STL) was applied, in addition to rigorous treatment for unit root removal and assurance of stationarity across all series.

3.1 Variable Panel

The empirical panel is composed of the time series detailed in Table 1:

Table 1: **Empirical Panel of Macro-Financial Variables**

Variable	Periodicity	Source
Bank Spread - Individuals	Monthly	BCB
Bank Spread - Corporate	Monthly	BCB
System Default Rate – Total	Monthly	BCB
SELIC Over Rate	Daily[1]	BCB
Brazil 5Y CDS (Country-Risk)	Daily[1]	Investing
Yield Curve Slope	Daily[1]	Investing[2]
Global VIX Index	Monthly	CBOE
Realized Exchange Rate Volatility	Daily[1]	BCB
B3 Financial Volume	Daily[1]	BOVESPA
Ibovespa Monthly Return	Daily[1]	BOVESPA
Economic Confidence Index (ICE) – Benchmark	Monthly	FGV

[1] Treatment applied for converting daily series to monthly frequency.
[2] Yield Curve Slope is defined by the difference between the 5-year CDS and the 1-year CDS.

3.2 Identification of Optimal Lags

Following the definition of the variable panel, we proceeded to identify the optimal lag for each series, taking the Economic Confidence Index (ICE) as a reference. The choice of the ICE stems from its aggregating nature. The indicator synthesizes the Confidence Surveys of **Manufacturing Industry**, **Commerce**, **Services**, and **Construction**, providing an overall view of domestic economic activity. Thus, it was used as a *proxy* for the current state of the Brazilian economy to guide the identification of time lags between financial indicators and the economic cycle.

Table 2 specifies the selected variables and their respective optimal lags:

Table 2: **Optimal Lag Structure by Variable Relative to the ICE**

Variable	Optimal Lag (in months)
Bank Spread - Individuals	1
Bank Spread - Corporate	1
System Default Rate – Total	1
SELIC Over Rate	7
Brazil 5Y CDS (Country-Risk)	1
Yield Curve Slope	10
Global VIX Index	2
Realized Exchange Rate Volatility	1
B3 Financial Volume	7
Ibovespa Monthly Return	2

4 Construction of the VIX Tupiniquim

With the data panel duly treated, stationary, and adjusted by optimal lags, we proceed to the quantitative modeling stage to extract the systemic stress thermometer. To this end, this section is divided into two complementary methodological approaches: informational condensation through Principal Component Analysis (PCA) and predictive filtering via Linear Regularization (LASSO, Ridge, and Elastic Net). For the construction of each VIX, the analyzed period encompasses from November 2011 to April 2026, covering a sample of 162 monthly observations.

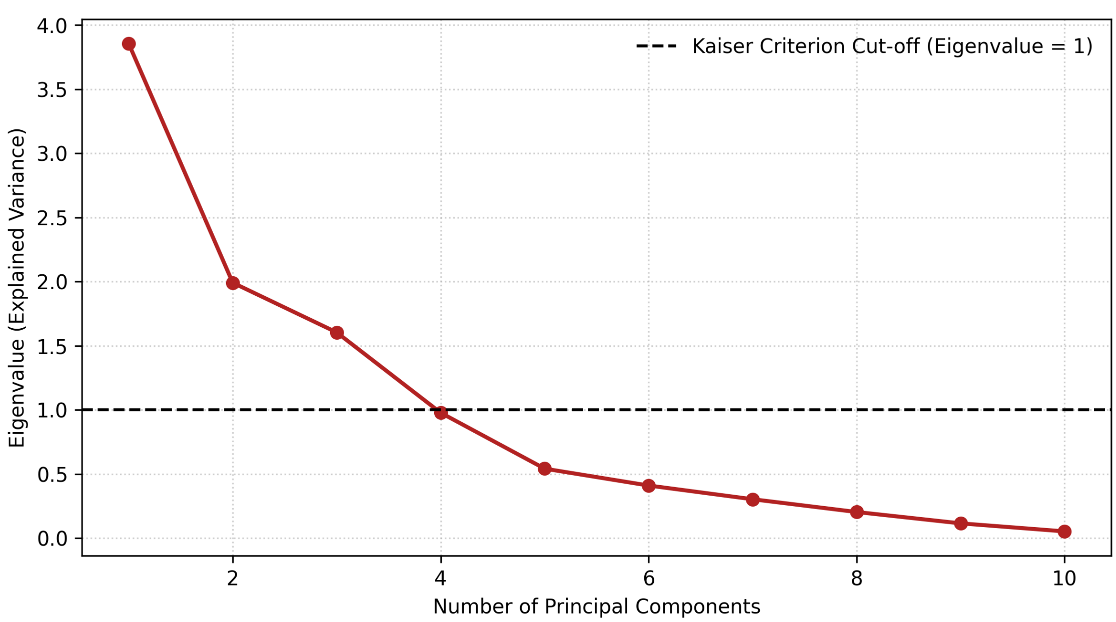
4.1 Principal Component Analysis (PCA)

The central objective of applying Principal Component Analysis (PCA) is to synthesize the volatility and co-movement of the sample's various time series into mutually orthogonal synthetic axes ($\approx 90^\circ$), reducing informational redundancy and multicollinearity. With the exception of the benchmark indicator — the Business Confidence Index (ICE), calculated and published by the Getulio Vargas Foundation (FGV) —, all standardized series participate in the covariance matrix that feeds the spectral decomposition algorithm.

To determine the ideal number of components capable of capturing stress dynamics without retaining residual stochastic noise, two traditional statistical criteria are employed: the *Kaiser Criterion* and the *Scree Plot*. The Kaiser criterion selects only components whose associated eigenvalues[1] are greater than 1, indicating that the retained latent factor conserves more variance than a single original standardized variable. Complementarily, the Scree Plot evaluates the rate of decay of marginal explained variance with each addition of an extra component.

Figure 1 illustrates the behavior of the eigenvalues and the cutoff line established by the model.

Figure 1: **Eigenvalues and cumulative explained variance by the PCA model**



The spectral decomposition algorithm identified 4 relevant principal components (eigenvalues >

[1] Eigenvalues are numbers that indicate how a linear transformation (represented by a matrix) "stretches" or "shrinks" a vector in a specific direction.

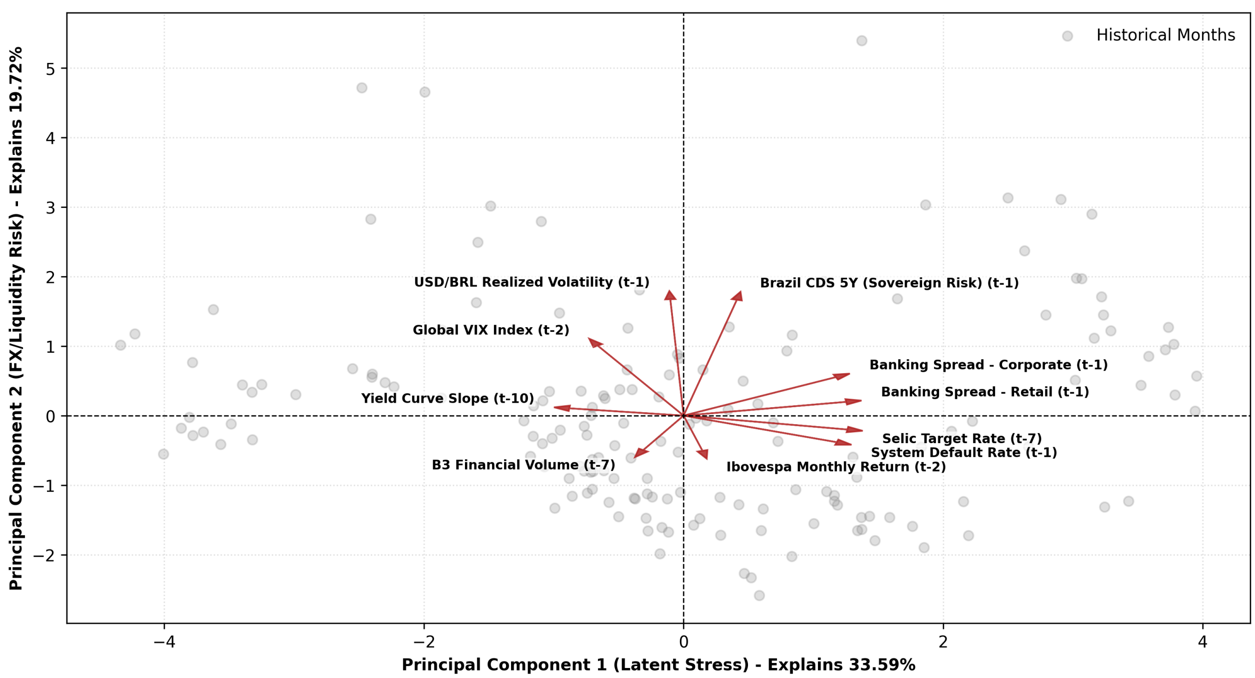
1), with PC_1 retaining the largest share of information from domestic and global macro-financial variables. Table 3 summarizes the states of each component and their respective explained variances.

Table 3: **Factorial Decomposition and Explained Variance by Principal Component**

Principal Component State (PC)	Explained Variance (%)
Latent Market Stress (PC_1)	33.59
Exchange Rate and Liquidity Risk (PC_2)	19.72
Cyclical Default Shocks (PC_3)	16.76
Residual Idiosyncratic Noise (PC_4)	10.64

Figure 2 reports the simultaneous projection of economic series and historical months onto the factorial plane composed of the first two principal components, which together concentrate **53.31% of the total system variance** (PC_1 with 33.59% and PC_2 with 19.72%).

Figure 2: **Factorial Biplot of Macroeconomic Forces (PC_1 vs. PC_2)**



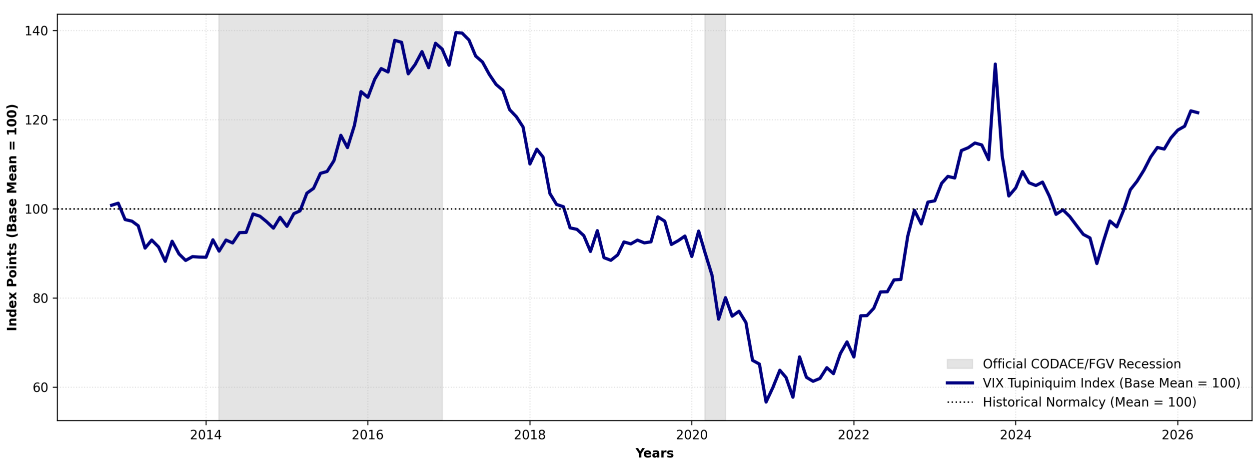
The **Principal Component 1 (PC_1)** functions as the main factor of domestic financial stress. It is observed that credit cost and credit risk variables — such as `Bank Spread - Corporate (t-1)`, `Bank Spread - Individuals (t-1)`, `Default Rate (t-1)`, and `SELIC Over Rate (t-7)` — project strongly in the positive direction of the semi-axis X. Periods located to the right of the plane represent moments of severe constraint in liquidity conditions and monetary tightening. In opposition ($\approx 180°$), the vector `Yield Curve Slope (t-10)` points to the left, demonstrating that the flattening or inversion of the yield curve accompanies cycles of rising capital costs.

The **Principal Component 2 (PC_2)** captures external volatility shocks and the repricing of public assets. The upper quadrant ($Y > 0$) allocates the vectors of `Brazil 5Y CDS (t-1)`, `Realized Exchange Rate Volatility (t-1)`, and `Global VIX Index (t-2)`. This set of variables indicates that jolts in country-risk perception and exchange rate volatility act as an orthogonal dimension to the purely domestic credit cycle, functioning as a second independent factor of macroeconomic instability.

The angular proximity between the `Spread Individuals` and `Spread Corporate` vectors confirms the strong co-movement of credit costs across segments. In turn, the opposition between `B3 Financial Volume (t-7)` (lower left quadrant) and the country-risk series (`CDS 5Y`) portrays the dynamics in which increases in country risk drain liquidity and traded volume in the domestic stock market.

The time trajectory of the resulting index, denoted as *VIX Tupiniquim* via PCA, is presented in Figure 3.

Figure 3: **Historical series of VIX Tupiniquim via Principal Components**



The behavior of the **VIX Tupiniquim via PCA** reveals important differences between the last two major recessive episodes in the Brazilian economy. During the 2014–2016 recession (Fiscal and Confidence Crisis), the indicator remained elevated for a prolonged period, with the persistence of the environment of uncertainty and economic deterioration being synchronous. As the economy began its recovery, the index gradually retreated, reflecting the normalization of macroeconomic conditions.

In contrast, in the 2020 recession, caused by the COVID-19 pandemic, the behavior was quite distinct. Although the shock was intense, its duration was very short (three months, according to CODACE[2]), causing the indicator to show a significant, yet transitory, peak. In other words, the index quickly captured the increase in uncertainty, but also its dissipation.

This contrast shows that the VIX Tupiniquim via Principal Components does not react solely to the intensity of economic shocks, but also to their persistence. The results suggest that the indicator is particularly effective for monitoring prolonged periods of economic stress, functioning as a synthetic measure of the evolution of uncertainty throughout the economic cycle.

4.2 Linear Regularization (LASSO, Ridge, and Elastic Net)

Even though Principal Component Analysis (PCA) is a highly effective methodology for synthesizing co-movements into orthogonal dimensions ($\approx 90^\circ$), it is characterized by unsupervised

[2] Economic Cycle Dating Committee, maintained by FGV.

learning — that is, without a target variable to guide it. To add another alternative in constructing the VIX, a supervised methodology was adopted, which, in addition to targeting the chosen benchmark (ICE), also exhibits a more parsimonious profile relative to confidence perceptions. For this purpose, the class of Linear Regularization models via coefficient penalization was utilized: *Least Absolute Shrinkage and Selection Operator* (LASSO), *Ridge Regression*, and *Elastic Net*.

These econometric approaches introduce a penalty technique to the Ordinary Least Squares (OLS) loss function, controlling overfitting and mitigating multicollinearity among macroeconomic variables. In the case of LASSO (L_1), it analyzes the variables and zeros out the weights of those that are redundant or irrelevant, building a sparse model. Ridge (L_2), unlike LASSO, does not zero out coefficients, but shrinks them toward zero. Finally, Elastic Net combines both penalization models (L_1 and L_2).

The calibration of the regularization parameter (α) for each model was performed via cross-validation on the training sample (80% of observations), preserving strict out-of-sample integrity. The selection of the winning model was based on the Bayesian Information Criterion (BIC) calculated on the test set (20% of data). Table 4 details the coefficients obtained after applying the three techniques.

Table 4: **Matrix of Penalized Coefficients by Regularization Model**

Variable (with lag)	**Coefficient LASSO (L_1)**	**Coefficient Ridge (L_2)**	**Coefficient Elastic Net**
Bank Spread - Individuals ($t-1$)	-2.0738	-1.4366	-1.3398
Bank Spread - Corporate ($t-1$)	0.0000	-1.4299	-1.5241
Default Rate ($t-1$)	0.9700	0.3965	0.2110
SELIC Over Rate ($t-7$)	-2.9060	-1.5674	-1.3508
Brazil 5Y CDS ($t-1$)	-3.7489	-3.2339	-3.1747
Curve Slope ($t-10$)	0.2269	0.6182	0.6073
Global VIX Index ($t-2$)	-2.7949	-2.2434	-2.0781
Exchange Volatility ($t-1$)	-0.8914	-1.2713	-1.3202
B3 Volume ($t-7$)	3.4296	2.1957	1.9719
Ibovespa Return ($t-2$)	0.4622	0.6558	0.6534

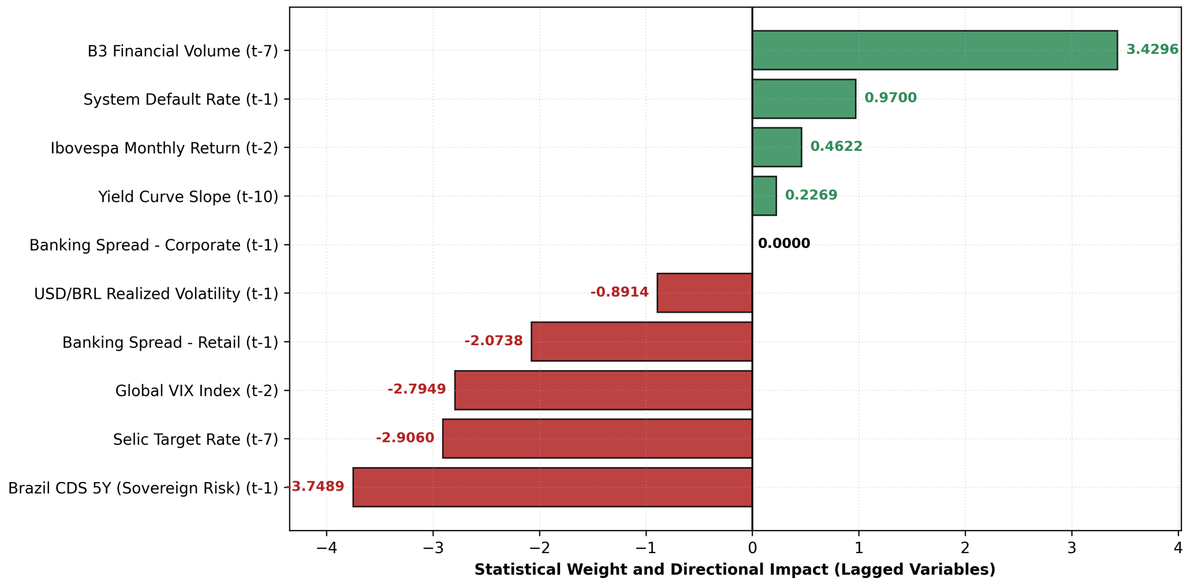
As shown in Table 5, the **LASSO** model presented the lowest BIC (145.0735) compared to the other models, displaying a more parsimonious configuration of the system with the active selection of 9 out of 10 lagged regressors.

Table 5: **Predictive Validation and Selection Matrix via BIC**

Regularized Model	**Active Coefficients (k)**	**BIC Criterion**
LASSO (L_1)	9	145.0735
Ridge (L_2)	10	146.1612
Elastic Net	10	145.8280

Figure 4 details the magnitude and direction of the standardized coefficients selected by the LASSO model.

Figure 4: **Weights and Direction of Statistical Impact of Variables (LASSO Model)**



The coefficient structure revealed by LASSO exposes the hierarchy of forces impacting risk perception and confidence in the domestic market:

1. **Country-Risk Anchor**: `Brazil 5Y CDS (t-1)` presented the largest penalized weight of a contractionary/stress nature (-3.7489), consolidating itself as the main leading variable for sovereign risk.

2. **Monetary Cycle and Credit Cost**: Variables associated with domestic monetary and banking tightening — `SELIC Over Rate (t-7)` (-2.9060) and `Spread Individuals (t-1)` (-2.0738) — maintained a strong negative impact, ratifying that the rising cost of capital restricts confidence.

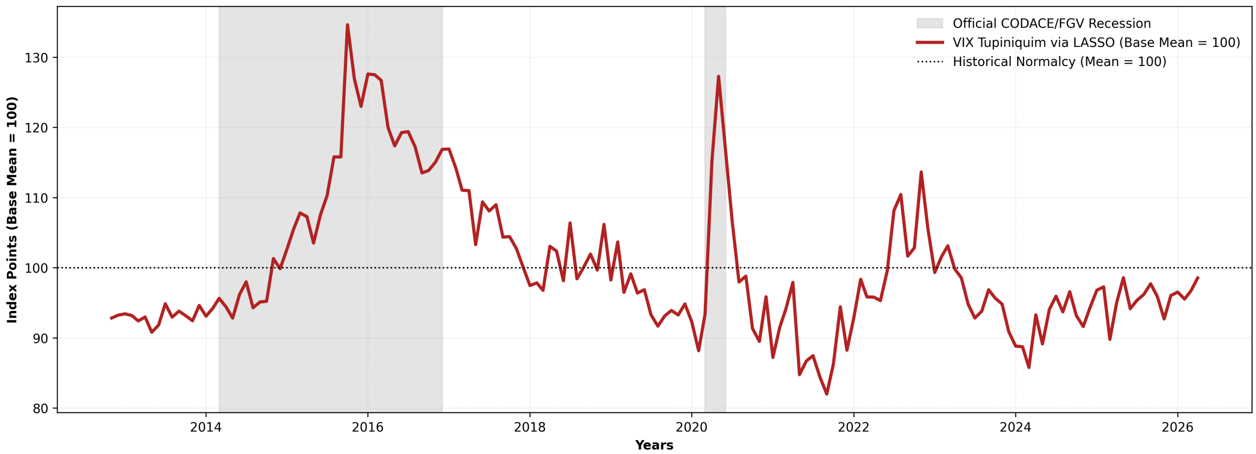
3. **Volatility and Exogenous Contagion**: `Global VIX Index (t-2)` (-2.7949) and `Realized Exchange Rate Volatility (t-1)` (-0.8914) act as direct vectors of contagion and external risk aversion.

4. **Liquidity and Market Activity**: `B3 Financial Volume (t-7)` ($+3.4296$) and `Default Rate (t-1)` ($+0.9700$) project with positive coefficients in the regularized structure of the model.

5. **Zeroed Variable (Expunged)**: Notably, the algorithm completely zeroed out the coefficient for `Spread Corporate (t-1)` (0.0000), eliminating the redundancy of this series relative to the `Spread Individuals` channel and sovereign risk.

Following the selection of coefficients estimated by the LASSO model, the standardized historical series of the **VIX Tupiniquim (via LASSO)** was constructed, presented in Figure 5.

Figure 5: **Historical Series of VIX Tupiniquim via LASSO**



The behavior of the **VIX Tupiniquim estimated via LASSO** presents a dynamic distinct from that observed in the version based on Principal Components. During the 2014–2016 recession (Fiscal and Confidence Crisis), the indicator reached high levels early in the recessive period, but began its normalization process even before the official end of the recession. This behavior indicates greater sensitivity to inflection points in economic conditions, anticipating the reduction of stress perceived by the indicators that compose the index.

In the 2020 recession, caused by the COVID-19 pandemic, the performance was particularly expressive. The indicator recorded an abrupt peak precisely during the most acute period of the crisis and quickly returned to levels near normal, accompanying the short nature of this recessive episode. Unlike the PCA version, LASSO managed to represent with greater precision the intensity and short duration of this exogenous shock.

Together, the results suggest that the LASSO-based approach produces an indicator that is more responsive to changes in the economic environment. Although it exhibits less persistence during prolonged recessions, its ability to capture abrupt events and inflection points represents an advantage for near-real-time monitoring applications.

While VIX Tupiniquim via PCA behaves as an indicator of the **stock of economic stress**, preserving memory of recessive periods, the LASSO version approaches a **flow** indicator, responding more rapidly to changes in macroeconomic conditions. That is, PCA favors cycle persistence, whereas LASSO emphasizes inflection points and abrupt shocks.

5 Practical Application with the EPU

In addition to constructing the VIX Tupiniquim through linear modeling (PCA and LASSO), we sought to evaluate its relationship with one of the main uncertainty indicators present in the literature: the *Economic Policy Uncertainty Index* (EPU), developed by *Baker, Bloom and Davis*[3].

The EPU is an uncertainty indicator based on the frequency count of terms such as “uncertain” or “uncertainty”, “economic” or “economy”, and political-economic terms, such as regulation, deficit, budget, tax, central bank, among others. The data source originates from the newspaper **Folha de São Paulo** and the historical series begins in 1991. While the EPU measures the frequency of news related to economic and political uncertainty, the VIX Tupiniquim seeks to reflect the stress

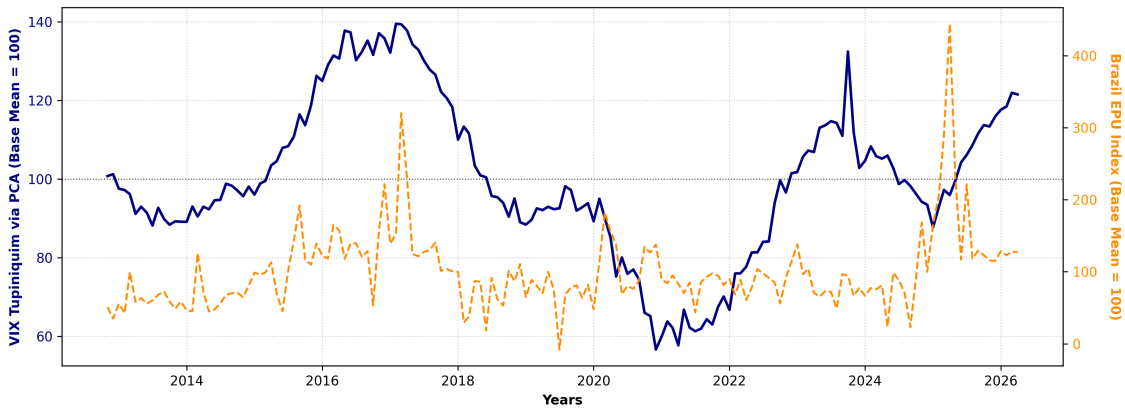
[3]For more information about the **Brazil EPU**, visit www.policyuncertainty.com

effectively priced by agents in the financial and credit markets, incorporating information from interest rates, exchange rates, spreads, and CDS.

Since the EPU series in levels presented a unit root according to the ADF test (p-value > 0.05), STL stochastic decomposition combined with first differencing (ΔEPU_{SA})[4] was applied to ensure the stationarity required for applying transmission tests.

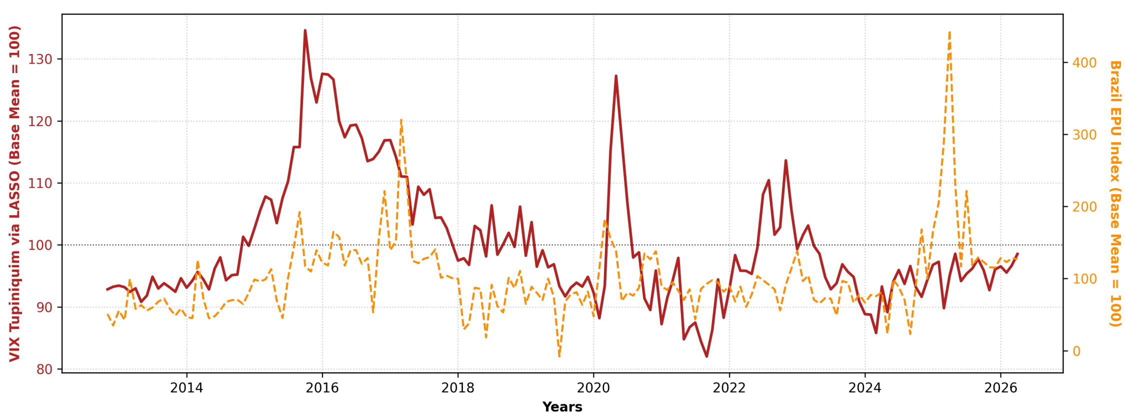
Figure 6 presents the historical comparison on a standardized scale (Average Base = 100) and dual axis between the VIX Tupiniquim via PCA and the EPU indicator.

Figure 6: **VIX Tupiniquim (PCA) vs. EPU Brazil**



In turn, Figure 7 illustrates the trajectory of the VIX Tupiniquim via LASSO in contrast to the EPU in the same visual format.

Figure 7: **VIX Tupiniquim (LASSO) vs. EPU Brazil**



The visual comparison of Figures 6 and 7 highlights relevant differences between the indicators. In several episodes of high political uncertainty reported by the press, the EPU registers significant peaks that do not find a corresponding match of equal magnitude in the VIX Tupiniquim. Consequently, the VIX Tupiniquim tends to respond primarily to episodes in which uncertainty translates into an effective deterioration of financial conditions.

To formally evaluate the temporal precedence between narrative uncertainty (ΔEPU) and market stress (VIX), Granger Causality tests were applied for lags of one to three months. Table 6 summarizes the F-statistics and their respective empirical p-values.

[4]The acronym SA indicates that the historical series underwent seasonal adjustment.

Table 6: **Granger Causality Test: VIX Tupiniquim vs. EPU Brazil**

Causality Relationship (H_0)	**Lag**	**F-Statistic**	**p-value**	**Conclusion (5% sig.)**
VIX (PCA) $\nrightarrow$ ΔEPU (VIX PCA does not cause News)	1	0.0000	0.9981	Does Not Reject H_0
	2	2.2453	0.1094	Does Not Reject H_0
	3	1.5848	0.1954	Does Not Reject H_0
ΔEPU $\nrightarrow$ VIX (PCA) (News does not cause VIX PCA)	1	0.3142	0.5759	Does Not Reject H_0
	2	0.6829	0.5067	Does Not Reject H_0
	3	0.4613	0.7097	Does Not Reject H_0
VIX (LASSO) $\nrightarrow$ ΔEPU (VIX LASSO does not cause News)	1	0.7750	0.3800	Does Not Reject H_0
	2	1.1607	0.3160	Does Not Reject H_0
	3	0.8149	0.4875	Does Not Reject H_0
ΔEPU $\nrightarrow$ VIX (LASSO) (News does not cause VIX LASSO)	1	1.2602	0.2633	Does Not Reject H_0
	2	1.4755	0.2319	Does Not Reject H_0
	3	1.6283	0.1852	Does Not Reject H_0

The results provided no sufficient statistical evidence to reject the null hypothesis of absence of Granger causality in any direction or lag considered: **there is a statistical dissociation between market risk restabilization and news-narrated uncertainty**.

Taken together, the visual evidence and econometric tests indicate that the VIX Tupiniquim captures a distinct dimension of economic uncertainty relative to the EPU. While the EPU reflects changes in the informational environment reported by the press, the VIX focuses on the risk perception effectively incorporated into financial asset prices. This distinction reinforces the potential of the indicator as a complementary tool for monitoring economic stress.